

07_FRONTEND.md

TASK 07

FRONTEND DEVELOPMENT

Status

Mandatory

Owner

Frontend Lead

Supporting Guides

Project Director

UI/UX Lead

Backend Lead

Security Architect

QA & Security Lead

Principal Engineer

Estimated Time

5 to 8 Days

Dependencies

00_PROJECT_CONTEXT.md

00_ENGINEERING_RULES.md

00_PROJECT_MEMORY.md

08_FRONTEND_LEAD.md

03_ARCHITECTURE.md

05_UI_DESIGN.md

06_BACKEND.md

Mission

Develop a complete, responsive, accessible and production-ready frontend for the Secure Dance Academy Management System.

Implement every approved design.

Integrate every backend API.

Maintain consistent user experience.

Support enterprise-quality usability.

Objectives

Implement all pages.

Build reusable components.

Connect backend APIs.

Support secure authentication.

Optimize performance.

Meet accessibility requirements.

Deliver production-ready code.

Development Principles

Follow

Clean Architecture

Component-Based Development

Feature-Based Structure

Responsive First

Accessibility First

Security by Default

Reuse before creating.

Never duplicate components.

Technology Stack

Framework

Next.js 15

Language

TypeScript

Styling

Tailwind CSS

Component Library

shadcn/ui

Forms

React Hook Form

Validation

Zod

Tables

TanStack Table

Charts

Recharts

Icons

Lucide React

Notifications

Sonner

Application Structure

Implement

Layouts

Protected Routes

Public Routes

Navigation

Shared Components

Feature Modules

Utilities

Hooks

Providers

Error Pages

Loading Components

Maintain feature isolation.

Authentication Screens

Implement

Login

Register

Forgot Password

Reset Password

Email Verification

Session Expired

Access Denied

Logout

Profile

Password Change

Integrate with Supabase Authentication.

Dashboard Pages

Administrator Dashboard

Coach Dashboard

Parent Dashboard

Artist Dashboard

Each dashboard should display

Statistics

Quick Actions

Notifications

Recent Activity

Charts

Schedules

Upcoming Events

Role-specific information only.

User Management Pages

Create User

Edit User

Delete User

User Details

Role Assignment

Search

Filtering

Pagination

Bulk Actions

Audit History

Artist Module

Implement

Artist List

Artist Details

Profile

Medical Information

Emergency Contacts

Performance History

Attendance History

Coach Assignment

Parent Assignment

Status

Photo Upload

Parent Module

Implement

Parent Dashboard

Children Overview

Attendance

Performance

Notifications

Medical Records

Profile

Parents should only view their own children.

Coach Module

Implement

Coach Dashboard

Assigned Artists

Attendance

Performance

Training Sessions

Medical Updates

Reports

Calendar

Attendance Module

Implement

Attendance List

Attendance Form

Attendance Calendar

Attendance Reports

Statistics

Search

Filters

Export

Duplicate Prevention

Performance Module

Implement

Performance List

Competition Results

Training Scores

Statistics

Achievements

Charts

Progress History

Reports

Injury Module

Implement

Injury List

Medical Notes

Recovery Status

History

Clearance

Restricted Access

Audit Timeline

Activities Module

Implement

Activity List

Calendar

Schedules

Participants

Registration

Attendance

History

Notifications

Reports Module

Implement

Attendance Reports

Performance Reports

Medical Reports

Coach Reports

Audit Reports

Export PDF

Export CSV

Filtering

Charts

Settings Module

Implement

Profile

Account

Password

Preferences

Notifications

Theme

Security Settings

Session Management

Reusable Components

Create

Buttons

Cards

Forms

Inputs

Select

Date Picker

Tables

Charts

Dialogs

Drawers

Badges

Avatar

Search

Filters

Pagination

Breadcrumb

Sidebar

Navbar

Footer

Loading Skeletons

Empty States

Error States

Confirmation Dialogs

Every component should be reusable.

Forms

Every form should include

Validation

Loading State

Error Handling

Success Messages

Confirmation Dialogs

Autosave where appropriate

Prevent duplicate submissions.

Navigation

Implement

Sidebar

Top Navigation

Breadcrumb

Quick Search

Notifications

User Menu

Responsive Mobile Navigation

Navigation should remain consistent.

State Management

Use

Server Components

Client Components only when necessary

Server Actions

React Hooks

Minimal Client State

Avoid unnecessary global state.

API Integration

Integrate every backend endpoint.

Handle

Loading

Success

Failure

Validation Errors

Unauthorized

Forbidden

Expired Session

Retry

Never expose backend errors directly.

Accessibility

Support

Keyboard Navigation

Screen Readers

ARIA Labels

Focus Management

Contrast

Responsive Typography

Semantic HTML

Meet WCAG 2.2 AA.

Performance

Optimize

Images

Fonts

Bundle Size

Lazy Loading

Code Splitting

Caching

Streaming

Reduce unnecessary rendering.

Security

Never expose secrets.

Never trust client validation.

Hide unauthorized actions.

Respect RBAC.

Escape dynamic content.

Display safe error messages.

Support secure sessions.

Responsive Design

Support

Mobile

Tablet

Laptop

Desktop

Ultra Wide

No feature should be unavailable on smaller devices.

Error Handling

Implement

404

403

500

Offline Mode

Validation Errors

API Errors

Network Errors

Unexpected Errors

Every error should guide users toward recovery.

Documentation

Document

Components

Pages

Props

Hooks

Layouts

API Integration

Accessibility

Performance

Developer Notes

Deliverables

Complete Frontend

Role Dashboards

Responsive Pages

Authentication

CRUD Interfaces

Forms

Tables

Charts

Navigation

Component Library

API Integration

Accessibility Report

Performance Review

Frontend Documentation

Acceptance Criteria

All pages are implemented.

All APIs are integrated.

Responsive layouts work.

Accessibility passes.

Performance targets are met.

Components are reusable.

Navigation is complete.

Security guidelines are followed.

Documentation is complete.

Update Project Memory

Mark

Frontend Development

Completed

Record

Pages Completed

Components Created

API Integrations

Accessibility Status

Performance Improvements

Known Issues

Next Task

Security Implementation

Frontend Review Checklist

Verify

Pages render correctly.

Authentication works.

Navigation works.

Forms validate.

Tables function.

Charts display correctly.

Accessibility passes.

Performance is optimized.

Responsive layouts work.

No critical UI defects remain.

Documentation is complete.

Definition of Done

The frontend is production ready.

Every approved design has been implemented.

Every backend feature is integrated.

Every page is responsive.

Accessibility requirements are satisfied.

Performance is optimized.

Security best practices are followed.

The frontend is approved for security testing and final integration.